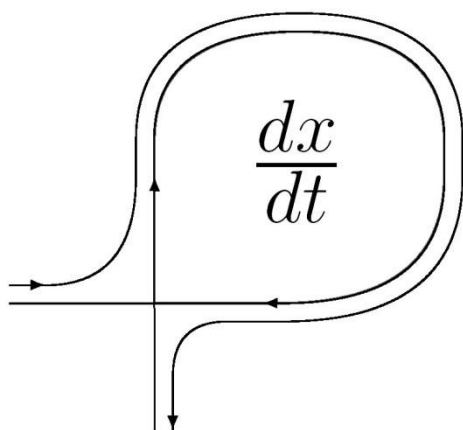




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Optimal control

## Trajectory Stabilization of Linear Systems with Jump Drift under Discrete Control Constraints

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**Abstract.** For a standard state stabilization problem of a linear stochastic differential system whose drift evolves in a direction determined by an independent Markov chain, a formulation with integer-valued constraints on the admissible control inputs is investigated. The stabilization objective is specified by a general quadratic functional. Cases with full information (the state of the chain is known) and with indirect observations (the system state serves as an indirect observation of the unknown chain state) are considered. In contrast to the solved unconstrained problem, the conditions for the existence of an optimal solution are not satisfied for the “integer-valued” formulation. The task is to find and implement an  $\varepsilon$ -optimal control. It is shown that such a control is obtained by discretizing the optimal solution of the unconstrained problem and applying mixed-integer nonlinear programming. The implementation of  $\varepsilon$ -optimal control, due to the stochastic nature of the problem and the possibility of an unbounded number of switchings, may require excessive computational effort. A practically implementable approximation is obtained by a relaxation method: it is computed as the result of an integer-valued transformation of the optimal control in the unconstrained problem. Three variants of such transformations are proposed. A numerical experiment based on a model of the dynamic evolution of the position of a simple mechanical actuator is carried out. The computational results confirm the applicability of the proposed solutions with respect to the stabilization objective and also allow a comparison of relaxation strategies depending on the “cost” of the control input.

**Keywords:** linear system stabilization; quadratic performance functional; dynamic programming; feedback control; Wonham filter; mixed-integer nonlinear programming; relaxation method; mechanical drive.

## 1. Introduction

A large number of problems in scientific, technical, engineering, financial, and even humanitarian applications can be reduced to formulations of optimal control of the state of a dynamical system. Stochastic models describing uncertainties of a random nature possess certain technical features related to differentiation and the interpretation of boundedness [1,2]. In many formulations, these nuances, when searching for an effective solution method, lead to dynamic programming and the Bellman equation, the practical solution of which, as a rule, is possible only in the absence of constraints on the values of the control input. Constraints on the values of control inputs may include requirements on the finiteness of second moments of random functions, including the control itself; however, formal geometric constraints characteristic of deterministic formulations are usually absent. There exists, however, a small but deserving-of-separate-attention class of problems that has special properties in this respect. These are problems in which there are constraints on the values of the control inputs consisting in requirements of their discreteness, that is, requirements of finiteness or countability of the set of admissible controls [3].

In typical optimization formulations, methods of mixed-integer nonlinear programming [4] (mixed-integer nonlinear programming, MINP) are used quite successfully to solve problems with discrete, more precisely, “integer-valued,” constraints. These methods combine intelligent enumeration in spaces with integer constraints, and therefore have combinatorial complexity, with methods of nonlinear optimization. In the control case, these include the maximum principle [5] (traditionally more effective in deterministic formulations) and the already mentioned dynamic programming [6] (more successful in stochastic formulations). In general, problems in which the constraints on the optimal control consist in the fact that the state, measurements, and/or control take integer values become significantly more complex. These are mixed problems that involve both optimal control and integer optimization.

It should be noted that, from a formal point of view, the mixed type of problems does not introduce anything fundamentally new. The questions that are usual for optimal control remain in the foreground: whether the problem is convex and whether optimality criteria can be applied; if the problem is nonconvex, by what methods it can be made convex and transformed into a solvable optimization problem. Issues of integer optimization arise in the background; they are not of a fundamental nature, and their complexity is rather computational. Integer optimization is the selection from a finite or countable set of alternatives, that is, enumeration and NP-hard problems. When the number of alternatives is large and complete enumeration is impossible, numerous and highly diverse heuristic approaches are successfully applied, allowing infeasible optimal solutions of complete enumeration to be transformed into the domain of searching for acceptable, practically implementable approximations. General information on this topic is well reflected in the survey [3].

The MINP problem addressed in this paper belongs to the field of stochastic optimal control. The application of integer programming in this area is not very typical. More often, researchers solve deterministic problems in which the optimal control and the optimal trajectory exist and are specified, or at least approximated, by ordinary functions rather than probability distributions. The control problem under consideration consists in stabilizing a linear system about a trajectory generated by a Markov chain, that is, drifting in jumpwise changing directions. In addition, the system state is affected by additive Gaussian noise, so that the target position must also be estimated from an indirect noisy measurement. The solution is provided by 1) a quadratic functional defining the stabilization objective as the mean square deviation of the system state vector from the target Markov trajectory, 2) the Wonham filter [7], which forms an optimal estimate of the chain, and 3) the separation principle, which reduces the problem to a full-information formulation and to the solution of the Bellman equation. This solution has been studied in detail in [8–10], and in this paper an additional integer-valued constraint is imposed on the control. We note the logical consistency of such a formulation: the integer-valued nature is dictated by the target trajectory—a piecewise-constant function taking a finite number of values “assigned” by the state index of the discrete Markov chain. Since the solution without integer constraints is available, the questions under study include the fundamental possibility of stabilizing the system with integer-valued control, the existence of a solution, a comparison of its efficiency with control that does not have such constraints, and methods for the practical implementation of the stabilization strategy.

Due to the stochastic nature of the problem, the use of most MINP methods does not appear to be feasible. The only practically promising approach turns out to be the relaxation method (relaxing integrality). Its idea is to first abandon the integer constraints and solve the optimization problem over a broader class, and then adapt the obtained solution to the integer constraints [11,12]. This approach is also advantageous in that it allows one to very conveniently assess possible losses due to approximation, since a “lower” bound is obtained automatically when solving the optimization problem with removed (relaxed) constraints.

Concluding the introductory part, we note that the idea of using integer constraints is not new and is indeed characteristic of a fairly wide range of applications. A large field of linear integer programming should be mentioned separately. This special, very broad class of problems has no direct relation to the subject of the present paper. For nonlinear models, one of the oldest examples of MINP belongs to chemical engineering and models the design of a multiproduct batch plant [13]. The most well-known example involving dynamics is the control of metro operation [14,15]. Its importance lies in the fact that it is a control problem that is dynamic and nonlinear. In this problem there are differential equations and classical optimization by means of the calculus of variations. Similar examples can readily be found in a wide variety of fields, for example, in the design of water supply systems [16–18] and gas supply systems [19–21], in the control of ventilation systems [22,23], and of power grids [24,25]. These examples show that the most obvious application area of MINP models is the control of resource distribution in systems. Indeed, there are also results in the fields of communication technology and computer science, for example, response to cyberattacks [26,27] and wireless network bandwidth allocation [28–30]. In surveys on MINP optimization [31–33], it is easy to find practical examples from such areas as nuclear power engineering, robotics, traffic optimization, nanomaterials and devices, biology, emergency situations, automotive engineering, finance, and many others.

## 2. Stabilization Problem with Integer-Valued Control Constraints

### 2.1. Problem Statement

On a canonical probability space  $(\Omega, \mathcal{F}, \mathcal{P}, \mathcal{F}_t)$ ,  $t \in [0, T]$ , consider a linear stochastic differential system with state  $z_t \in \mathbb{R}^{n_z}$ :

$$dz_t = a_t y_t dt + b_t z_t dt + c_t u_t dt + \sigma_t dw_t, z_0 = Z, \quad (1)$$

where  $y_t \in \mathbb{R}^{n_y}$  is a Markov chain with a finite number of states taking values in the set of unit coordinate vectors  $\{e_1, \dots, e_{n_y}\}$  in  $\mathbb{R}^{n_y}$ ;  $u_t \in \mathbb{Z}^{n_u}$  is the control; the disturbance  $w_t \in \mathbb{R}^{n_w}$  is a standard vector-valued Wiener process; the initial condition  $Z \in \mathbb{R}^{n_z}$  is a Gaussian random vector with known mean and covariance; the time-dependent functions  $a_t \in \mathbb{R}^{n_z \times n_y}$ ,  $b_t \in \mathbb{R}^{n_z \times n_z}$ ,  $c_t \in \mathbb{R}^{n_z \times n_u}$ , and  $\sigma_t \in \mathbb{R}^{n_z \times n_w}$ . The distribution  $\pi_0$  of the initial state  $y_0 = Y$  of the chain  $y_t$  and the transition rate matrix  $\Lambda_t$  are assumed to be known. By  $\mathbb{R}^n$  we denote the space of  $n$ -dimensional vectors with real entries, and by  $\mathbb{Z}^n$  the space of  $n$ -dimensional vectors with integer entries.

The control  $u_t$  generates trajectories  $z_t$  drifting in variable directions determined by the Markov jump process  $y_t$ . The formal stabilization objective consists in minimizing the cost functional

$$J(U_0^T) = \mathbb{E} \left\{ \int_0^T \|P_t y_t + Q_t z_t + R_t u_t\|_{S_t}^2 dt + \|P_T y_T + Q_T z_T\|_{S_T}^2 \right\}, \quad (2)$$

where  $U_0^T = \{u_t = U_t(z_t), 0 \leq t \leq T\}$ ,  $P_t \in \mathbb{R}^{n_J \times n_y}$ ,  $Q_t \in \mathbb{R}^{n_J \times n_z}$ ,  $R_t \in \mathbb{R}^{n_J \times n_u}$ ,  $S_t \in \mathbb{R}^{n_J \times n_J}$ ,  $S_t \geq 0$ ,  $S_t = S'_t$ ,  $0 \leq t \leq T$ , are given matrix-valued functions; the weighting function is defined as  $\|x\|_S^2 = x' S x$  for a symmetric positive semidefinite matrix  $S$ ; for the identity matrix  $S = E$ , the

Euclidean norm corresponds to  $\|x\|_E^2 = |x|^2$ ;  $x'$  denotes the transpose of the matrix (vector)  $x$ ;  $\mathbb{E}\{x\}$  denotes the mathematical expectation of the random vector  $x$ .

In addition to the fact that the class of admissible controls is defined as the class of state-feedback controls with respect to  $z_t$  [34], it is assumed that the random functions  $u_t$  have finite second moments. The assumptions that  $w_t, y_t, Y, Z$  are jointly independent, and that the functions  $\Lambda_t, a_t, b_t, c_t, \sigma_t$  are piecewise continuous, guarantee the existence of a unique strong (pathwise) solution of (1). Finally, the optimality conditions for the control also include the piecewise continuity of the functions  $P_t, Q_t, R_t, S_t$ , the nondegeneracy condition  $R'_t S_t R_t > 0$ , and  $\sigma_t \sigma'_t > 0$ .

## 2.2. Optimal Control without Integer-Valued Constraints

Note that the problem is formulated for the case of control based on indirect observations,  $u_t = U_t(z_t)$ . Such a feedback implies that the state of the chain  $y_t$  is unknown, while the system state  $z_t$  serves as an indirect observation.

This problem without integer-valued constraints on the control, i.e., for  $u_t \in \mathbb{R}^{n_u}$ , as well as the complete information case (admissible control with full feedback  $u_t = U_t(y_t, z_t)$ ) and the separation principle, have been studied in [8–10]. It is known that the optimal control loop  $u_t^* = U_t(z_t^*)$  includes an estimate  $\hat{y}_t$  of the chain state based on indirect observations  $z_t$ , which is given by the Wonham filter [7]:

$$\begin{aligned} d\hat{y}_t &= \Lambda'_t \hat{y}_t dt + (\text{diag}(\hat{y}_t) - \hat{y}_t \hat{y}'_t) a'_t (\sigma_t \sigma'_t)^{-\frac{1}{2}} \times \\ &\times (\sigma_t \sigma'_t)^{-1/2} (dz_t - a_t \hat{y}_t dt - b_t z_t dt - c_t u_t dt), \hat{y}_0 = \mathbb{E}\{Y\}. \end{aligned} \quad (3)$$

The optimal control  $u_t^*$  itself has the form:

$$u_t^* = -\frac{1}{2} (R'_t S_t R_t)^{-1} (c'_t (2\alpha_t z_t^* + \beta_t \hat{y}_t) + 2R'_t S_t (P_t \hat{y}_t + Q_t z_t^*)), \quad (4)$$

$$\begin{aligned} \frac{d\alpha_t}{dt} - (M_t^\alpha \alpha_t + \alpha'_t (M_t^\alpha)') + N_t^\alpha - \alpha'_t c_t (R'_t S_t R_t)^{-1} c'_t \alpha_t &= 0, \alpha_T = Q'_T S_T Q_T, \\ \frac{d\beta_t}{dt} + \beta_t \Lambda'_t + M_t^\beta - N_t^\beta \beta_t &= 0, \beta_T = 2Q'_T S_T P_T, \\ \frac{\partial \gamma_t}{\partial t} + y' \Lambda_t \Lambda'_t + M_t^\gamma y &= 0, \gamma_T y = y' P'_T S_T P_T y, \\ \frac{d\delta_t}{dt} + \text{tr}\{\sigma'_t \alpha_t \sigma_t\} &= 0, \delta_T = 0, \end{aligned} \quad (5)$$

where the following notation is used:

$$\begin{aligned} M_t^\alpha &= Q'_t S_t R_t (R'_t S_t R_t)^{-1} c'_t, \quad N_t^\alpha = Q'_t (S_t - S_t R_t (R'_t S_t R_t)^{-1} R'_t S_t) Q_t, \\ M_t^\beta &= 2((a'_t - P'_t S_t R_t (R'_t S_t R_t)^{-1} c'_t) \alpha_t + P'_t (S_t - S_t R_t (R'_t S_t R_t)^{-1} R'_t S_t) Q_t), \\ N_t^\beta &= Q'_t S_t R_t (R'_t S_t R_t)^{-1} c'_t + \alpha_t c_t (R'_t S_t R_t)^{-1} c'_t, \\ M_t^\gamma y &= N_t^\gamma(y) = y' \beta'_t (a_t - c_t (R'_t S_t R_t)^{-1} R'_t S_t P_t) y + \\ &+ y' P'_t (S_t - S_t R_t (R'_t S_t R_t)^{-1} R'_t S_t) P_t y - 1/4 y' \beta'_t c_t (R'_t S_t R_t)^{-1} c'_t \beta_t y. \end{aligned}$$

The solutions  $\alpha_t, \beta_t, \gamma_t(y), \delta_t$  of equations (5) are required to determine the Bellman function:

$$V_t = V_t(z) = \inf_{u_t^T} \mathbb{E} \left\{ \int_t^T \|P_s y_s + Q_s z_s + R_s u_s\|_{S_s}^2 ds + \|P_T y_T + Q_T z_T\|_{S_T}^2 \right\}$$

in the form  $V_t = z' \alpha_t z + z' \beta_t \hat{y}_t + \gamma_t \hat{y}_t + \delta_t$ , so that along the optimal trajectory  $z_t^*$  the control cost is  $V_0 = Z' \alpha_0 Z + Z' \beta_0 \hat{y}_0 + \gamma_0 \hat{y}_0 + \delta_0$ . Thus, the lower bound of the functional (2) for controls  $u_t \in \mathbb{R}^{n_u}$  has the form:

$$J((U^*)_0^T) = \inf J(U_0^T) = \mathbb{E}\{V_0\}. \quad (6)$$

The obtained solution of the unconstrained problem,  $u_t \in \mathbb{R}^{n_u}$ , can now be used to study the solution of the stated problem with discrete constraints,  $u_t \in \mathbb{Z}^{n_u}$ .

### 2.3. Optimal and $\varepsilon$ -Optimal Solutions of the Integer-Valued Stabilization Problem

For similar integer-valued problems in traditional deterministic control models, the study of MINP applications began with a simplified case [14], in which  $u_t$  takes only two values (binary control), i.e.,  $u_t \in \{-1, 1\}$ , and with a prescribed finite number of “switchings” (a switching naturally refers to a moment when  $u_t$  changes its value). This restriction can be generalized by assuming that each component of the vector  $(u_t)_k \in \{-1, 1\}$ ,  $k = 1, \dots, n_u$ , is binary. Such a narrowing of the class of admissible controls, together with a convex quadratic criterion, allows one to assert that an optimal solution exists and, in many typical formulations, to find it using optimal control methods. There are numerous examples of optimal control problems in which the maximum principle is effectively applied, and from the constraint  $u(t) \in [-1, 1]$  it follows that the optimal control has the form  $u(t) = \text{sign}(f(t))$ . Thus, the optimal control becomes a sequence of switchings, and the integer constraints are satisfied automatically. The solution is reduced to finding the optimal sequence of switching times. Problems of this kind can be very diverse, up to the control of hybrid systems with variable dimension [35]. In addition, to find an approximate solution, discretization can be applied, reducing the problem to an enumeration of possible switching sequences and the use of heuristic MINP methods.

This situation of global optimality changes radically if the restriction on the number of switchings is abandoned. Even without a stochastic component, such an abandonment leads to singularity and to the nonexistence of an optimal solution. Qualitatively, the problem is characterized by the well-known Fuller phenomenon—a situation in which the optimal control switches an infinite number of times over a finite interval. Moreover, in [36] this problem is associated with a quadratic criterion similar to the one considered in (2). This is a well-known problem that continues to be studied and in itself deserves surveys [37]. The study of this phenomenon continues even in very modern applications, for example, in the control of underwater drones [38], but within deterministic models. It is clear that the presence of the Wiener process  $w_t$  in (1) only exacerbates the Fuller phenomenon. Therefore, the first conclusion regarding the stabilization problem under consideration is the absence of an optimal solution. That is, although  $\inf J(U_0^T)$  for controls  $u_t \in \mathbb{Z}^{n_u}$  exists, since the lower bound of  $J(U_0^T)$  is defined in (6) and the functional (2) is quadratic in  $u_t$ , there generally does not exist a control  $(U^*)_0^T$  for which the minimum is attained,  $J((U^*)_0^T) = \inf J(U_0^T)$ , over the class  $u_t \in \mathbb{Z}^{n_u}$ , at least within the class of integer-valued piecewise-constant functions.

One may then resort to a traditional approach by prescribing the number of switchings in advance. An optimal solution for such a problem does exist, but the original meaning of drifting along the Markov chain will be distorted. Therefore, the subsequent objective is to find an  $\varepsilon$ -optimal solution and to combat excessive switchings (“chattering”), which should lead to practically implementable controls. Such a solution with arbitrary accuracy can be obtained in the form of an approximation of the unconstrained optimal control (4) by simple uniform discretization:

$$u_t^{\varepsilon(C)} = u_t^*(t_i), t \in [t_i, t_{i+1}), \quad (7)$$

$$0 = t_1 < \dots < t_i < t_{i+1} < \dots < t_N = T, t_{i+1} - t_i = \delta, i = 0, \dots, N - 1.$$

The control  $u_t^{\varepsilon(C)}$  in (7) is a piecewise-constant approximation of the continuous (C) optimal control  $u_t^*$  for a uniform partition of the control interval  $[0, T]$  with step  $\delta$ . Since almost all trajectories of the optimal control  $u_t^*$  (4) are piecewise continuous, because the solutions of the ordinary differential equations (5) are piecewise continuous, by decreasing  $\delta$  one can ensure that the difference between the value (6) achieved by the optimal control  $u_t^*$  and the value of the cost functional (2) for the control  $u_t^{\varepsilon(C)}$  is smaller than any prescribed  $\varepsilon$ . This is evident for the case of control with full feedback. For control based on indirect observations, it is additionally necessary to represent the cost functional (2) as the result of the separation of the control and filtering problems:

$$J(U_0^T) = \mathbb{E} \left\{ \int_0^T \|P_t \hat{y}_t + Q_t z_t + R_t u_t\|_{S_t}^2 dt + \|P_T \hat{y}_T + Q_T z_T\|_{S_T}^2 + \int_0^T \|P_t (y_t - \hat{y}_t)\|_{S_t}^2 dt + \|P_T (y_T - \hat{y}_T)\|_{S_T}^2 \right\}. \quad (8)$$

The continuity of all functions appearing in (8) with respect to  $t$ , including those implicitly entering the solution  $z_t$ , and the independence of the estimate  $\hat{y}_t$ , determined by the Wonham filter (3), from any control, as well as of its accuracy  $y_t - \hat{y}_t$ , guarantee that by an appropriate choice of the partition step  $\delta$  the inequality  $|J((U^*)_0^T) - J((U^{\varepsilon(C)})_0^T)| < \varepsilon$  can be ensured.

Now, in order to prove the existence of an  $\varepsilon$ -optimal integer-valued (I) control  $u_t^{\varepsilon(I)}$ , two steps must be taken. First, it suffices to search for controls  $u_t^{\varepsilon(I)}$  among all controls of the form  $u_t$ :

$$(u_t)_k = \left[ \left( u_t^{\varepsilon(C)} \right)_k \right], \pm 1, \pm 2, \dots, \pm M, k = 1, \dots, n_u, t \in [t_i, t_{i+1}), \quad (9)$$

where  $[x]$  denotes the integer part of the number  $x$ . If the number  $M$  can be arbitrarily large, then the class of controls (9) will not differ from the class originally specified in the problem formulation. Therefore, the second step is to guarantee the finiteness of  $M$ . This is ensured by the existence of an “upper” bound—the value of the functional (2) under the zero control  $u_t = 0$ . Then, the existence of the sought  $\varepsilon$ -optimal solution follows from the finiteness of the number of variants in (9).

Thus, the control  $u_t^{\varepsilon(I)}$  is sought among piecewise-constant functions that, at each partition point  $t_i$ , are specified by replacing the control without integer-valued constraints by the nearest integer to the left or to the right for each coordinate, or by integers at a distance of  $\pm 1$ , and so on. At the same time, even if one restricts the optimization to controls that deviate from  $u_t^{\varepsilon(C)}$  by less than 1 in at least one coordinate (which, from the viewpoint of practical expediency, might be sufficient), the number of variants will be significant. Thus, the principal possibility of computing the control  $u_t^{\varepsilon(I)}$  by simple enumeration of all variants does not eliminate the issue of practical implementation. In this respect, some effective approximation by any heuristic MINP method is required.

### 3. Approximate Algorithms for $\varepsilon$ -Optimal Stabilization

Since, unlike deterministic models that describe a single trajectory for each admissible control, in the problem under consideration any combinatorial optimization must be performed for each possible realization of the output  $z_t$ , that is, for the trajectory corresponding to its probability distribution (together with the corresponding trajectory of the estimate  $\hat{y}_t$  and of the chain  $y_t$ ), this severely restricts the choice of applicable MINP methods. In essence, the only workable option is to construct heuristic approximations  $u_t^{\varepsilon(I)}$  by transforming  $u_t^{\varepsilon(C)}$  into a piecewise integer-valued control.



This idea of using a solution of a control problem obtained over a broader class of admissible controls has long been known as the relaxation method (relaxed control problem) [11,12]. The simplest variant of this method in the problem under consideration yields the following control  $u_t^{I_1}$ :

$$(u_t^{I_1})_k = \text{round} \left( (u_t^{\varepsilon(C)})_k \right), k = 1, \dots, n_u, \quad (10)$$

where  $\text{round}(x)$  denotes the integer rounding function of the number  $x$ . Such a control follows the standard rounding strategy [3]. Its drawback is the lack of suppression of the Fuller effect: each time the unconstrained control crosses the midpoint of a certain integer interval, a switching is performed in the integer-valued control. If, in the full-information formulation, one may assume that switchings (changes of the states of the chain  $y_t$ ) lead to impulsive changes of the output, which automatically eliminate “chattering,” this is not the case in the formulation with indirect observations. The Wonham filter (3), or its approximation, while counteracting observation errors generated by the Wiener process  $w_t$ , and reacting to changes in the state  $y_t$  even under very accurate observations, has a certain period during which the estimate exhibits transient behavior in response to the jumpwise change of  $y_t$ . Following the estimate, the strategy (10) will also oscillate. To address this drawback, two techniques are proposed [3]: sum of rounding and sum of rounding with a different threshold. Adapted to the problem under consideration, these strategies take the form:

$$(u_t^{I_2})_k = \text{round} \left( (u_t^{\varepsilon(C)})_k \right), \quad \text{если} \quad \left| (u_{t-l}^{\varepsilon(C)})_k - \text{round} \left( (u_t^{\varepsilon(C)})_k \right) \right| + \sum_{l=0}^{L-2} \left| (u_{t-l}^{\varepsilon(C)})_k - (u_{t-l}^{I_2})_k \right| \geq \frac{L}{2}, \quad (11)$$

$$(u_t^{I_2})_k = (u_{t-1}^{I_2})_k, \quad \text{иначе; } k = 1, \dots, n_u,$$

$$(u_t^{I_3})_k = \text{round} \left( (u_t^{\varepsilon(D)})_k \right), \quad \text{если} \quad \lambda_{l-1} \left| (u_{t-l}^{\varepsilon(C)})_k - \text{round} \left( (u_t^{\varepsilon(C)})_k \right) \right| + \sum_{l=0}^{L-2} \lambda_l \left| (u_{t-l}^{\varepsilon(C)})_k - (u_{t-l}^{I_3})_k \right| \geq \frac{1}{2}, \quad (12)$$

$$(u_t^{I_3})_k = (u_{t-1}^{I_3})_k, \quad \text{иначе; } k = 1, \dots, n_u.$$

Both controls  $u_t^{I_2}$  in (11) and  $u_t^{I_3}$  in (12) implement the same strategy, which consists in the following: the integer-valued control at time step  $t$  for the  $k$ -th coordinate is changed (a switching from the previous value  $(u_{t-1}^{I_{2/3}})_k$  to the new value  $(u_t^{I_{2/3}})_k$ , obtained by rounding the unconstrained optimal control) only if the necessity of switching has been confirmed over the last  $L$  steps (time instants). In the control  $u_t^{I_2}$  in (11), it is required that the difference between the unconstrained control and the rounded one be greater than 0.5 by a factor of  $L$ , including the current step. The control  $u_t^{I_3}$  in (12) introduces weights in order to assign greater significance to less distant time instants: it is assumed that  $\sum_{l=0}^{L-1} \lambda_l = 1$ ,  $\lambda_l > \lambda_{l-1} > 0$ . In this way, a less inert control is realized, which should be closer to variant (10).

## 4. Numerical Experiment

### 4.1. Mechanical Actuator Model and Baseline Computation

To carry out the practical experiment, we use a gantry crane model describing the motion of a trolley for lifting a load along a horizontal rail [8–10]. Giving meaning to integer-valued control and demonstrating its difference from continuous strategies can be achieved through penalties on the magnitude of the control input and through the choice of gain coefficients that scale the control to the range of the output variable. Thus, let the position of the trolley relative to the initial (extreme) position be described by the scalar variable  $X_t$ . The trolley tends to reach this position with velocity  $V_t$ . The magnitude and direction of the velocity are determined by a controlled force that depends linearly on the current position, the current velocity, the uncontrolled jumpwise input  $y_t$ , and the control  $u_t$ . The Markov chain  $y_t$  in this example models a sequence of indices of tracks occupied by arriving transport for loading. Altogether:

$$\begin{aligned} dX_t &= V_t dt, \quad t \in (0, T], \\ dV_t &= AX_t dt + BV_t dt + Cy_t dt + Hu_t dt + \sqrt{G} dW_t. \end{aligned} \quad (13)$$

The chain  $y_t$  has three states, the transition rate matrix  $\Lambda_t = \Lambda = \begin{pmatrix} -0,5 & 0,5 & 0 \\ 0,5 & -1 & 0,5 \\ 0 & 0,5 & -0,5 \end{pmatrix}$ , and the initial distribution  $\pi_0 = (1, 0, 0)'$ . Model (13) can be reduced to the form (1) by introducing the notation  $z_t = (X_t, V_t)'$ ,  $a_t = C = (C_1, C_2, C_3)$ ,  $b_t = \begin{pmatrix} 0 & 1 \\ A & B \end{pmatrix}$ ,  $c_t = H$ ,  $w_t = (0, W_t)'$ ,  $\sigma_t = (0, \sqrt{G})$ . In this case, the nondegeneracy condition for the noises in the Wonham filter (3) is formally violated. This circumstance can be readily taken into account by using only  $dV_t$  to compute the estimate  $\hat{y}_t$ . This is admissible since  $dX_t$  is expressed linearly in terms of  $dV_t$ . The initial conditions  $X_0$  and  $V_0$  are assumed to be independent standard Gaussian random variables. In the computations, the model parameters are chosen as follows:

$$A = -0,25; B = -0,5; G = 0,01; C_1 = -1,5; C_2 = -0,5; C_3 = 0,5; H = 100; T = 10.$$

The control objective consists in tracking, by the coordinate  $X_t$ , the position  $Dy_t$ , where  $D = (-6, -2, 2)$ , and is formalized by the cost functional

$$J(U_0^T) = \mathbb{E} \left\{ \int_0^T (|Dy_t - X_t|^2 + R|u_t|^2) dt \right\}. \quad (14)$$

The discretization was performed with step  $\delta = 0.001$ . To estimate the value of the cost functional, 1000 trajectories of (13) were simulated, over which averaging was carried out to obtain an estimate of the quantity (14) for different control strategies. For convenience, all control strategies mentioned are listed in Tab. 1, including  $u_t^{\varepsilon(I)}$ . Computing these controls, even in such a simple experiment, is not feasible; however, it is clear that the strategies  $u_t^{\varepsilon(C)}$  and  $u_t^{I_{1/2/3}}$  provide lower and upper bounds for the value of the cost functional (14) corresponding to  $u_t^{\varepsilon(I)}$ . This qualitative assessment is the limit to which we must restrict ourselves.



Tab 1. Notation for stabilization strategies

| Model type                                      | Control strategy                  |                                       |                          |                        |                                                   |
|-------------------------------------------------|-----------------------------------|---------------------------------------|--------------------------|------------------------|---------------------------------------------------|
|                                                 | $\varepsilon$ -optimal continuous | $\varepsilon$ -optimal integer-valued | Integer-valued heuristic |                        |                                                   |
|                                                 |                                   |                                       | standard rounding, $I_1$ | sum of rounding, $I_2$ | sum of rounding with a different threshold, $I_3$ |
| <b>Complete information,</b><br>$y_t \hat{y}_t$ | $U_{y_t}^{\varepsilon(C)}$        | $U_{y_t}^{\varepsilon(I)}$            | $U_{y_t}^{I_1}$          | $U_{y_t}^{I_2}$        | $U_{y_t}^{I_3}$                                   |
| <b>Indirect observations,</b><br>$\hat{y}_t$    | $U_{\hat{y}_t}^{\varepsilon(C)}$  | $U_{\hat{y}_t}^{\varepsilon(I)}$      | $U_{\hat{y}_t}^{I_1}$    | $U_{\hat{y}_t}^{I_2}$  | $U_{\hat{y}_t}^{I_3}$                             |

Finally, we note that in computing the controls for models with indirect observations, instead of the Wonham filter estimate  $\hat{y}_t$  (3), its stable numerical approximation [39] was used. In the full-information variants,  $\hat{y}_t$  was not computed and was replaced by the exact value  $y_t$ . Finally, for the “sum of rounding” strategies, the rounding interval was chosen equal to 3, and the weights for the “sum of rounding with a different threshold” strategy were set to 0,5; 0,33; 0,17.

The first computation, referred to as the baseline case, was performed for the value  $R = 0.1$ . The results are presented in Tab. 2.

Table 2. Control performance in the baseline model with penalty  $R = 0.1$ 

| $J(U_0^T)$ | $J\left(\left(U_{y_t}^{\varepsilon(C)}\right)_0^T\right)$ | $J\left(\left(U_{y_t}^{I_1}\right)_0^T\right)$ | $J\left(\left(U_{y_t}^{I_2}\right)_0^T\right)$ | $J\left(\left(U_{y_t}^{I_3}\right)_0^T\right)$ | $J\left(\left(U_{\hat{y}_t}^{\varepsilon(C)}\right)_0^T\right)$ | $J\left(\left(U_{\hat{y}_t}^{I_1}\right)_0^T\right)$ | $J\left(\left(U_{\hat{y}_t}^{I_2}\right)_0^T\right)$ | $J\left(\left(U_{\hat{y}_t}^{I_3}\right)_0^T\right)$ |
|------------|-----------------------------------------------------------|------------------------------------------------|------------------------------------------------|------------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------|------------------------------------------------------|------------------------------------------------------|
|            | 10,88                                                     | 11,15                                          | 11,30                                          | 11,33                                          | 17,96                                                           | 18,21                                                | 18,19                                                | 18,25                                                |

Graphical illustrations for this example are presented in the following section. Here we note that the range of the obtained integer-valued control values spans from  $-15$  to  $25$  and, judging by the values of the cost functional in Table 2, there is no significant difference between the continuous and integer-valued control strategies. On the one hand, this computation shows that, under integer-valued constraints, stabilization can be achieved as effectively as with unconstrained strategies. In particular, it confirms the thesis that there is no practical necessity to compute the “exact” strategy  $u_t^{(\varepsilon(I))}$ . On the other hand, with such a range of control magnitudes, especially taking into account the scaling coefficient  $H = 100$ , it is not very convenient to analyze the differences between strategies, their quality, and the nature of switchings. Therefore, the purpose of the next section is to present experimental results obtained by achieving a substantial narrowing of the range of integer-valued control magnitudes.

## 4.2. Analysis of Control Strategies with Increasing Penalty for Integer-Valued Control

In this section, three experiments with  $R = 1$ ,  $R = 2.5$ , and  $R = 5$  are added to the baseline model. All other parameters, including the set of disturbance trajectories  $W_t$  and the initial conditions  $X_0$  and  $V_0$ ,

are kept unchanged. The results of all computations are summarized in Tab. 3; the result of Tab. 1 is also reproduced.

Table 3. Control performance in models with different penalties

| $J(U_0^T)$ | $J((U_{y_t}^{\varepsilon(C)})_0^T)$ | $J((U_{y_t}^{I_1})_0^T)$ | $J((U_{y_t}^{I_2})_0^T)$ | $J((U_{y_t}^{I_3})_0^T)$ | $J((U_{\hat{y}_t}^{\varepsilon(C)})_0^T)$ | $J((U_{\hat{y}_t}^{I_1})_0^T)$ | $J((U_{\hat{y}_t}^{I_2})_0^T)$ | $J((U_{\hat{y}_t}^{I_3})_0^T)$ |
|------------|-------------------------------------|--------------------------|--------------------------|--------------------------|-------------------------------------------|--------------------------------|--------------------------------|--------------------------------|
| $R = 0,1$  | 10,88                               | 11,15                    | 11,30                    | 11,33                    | 17,96                                     | 18,21                          | 18,19                          | 18,25                          |
| $R = 1$    | 18,73                               | 21,85                    | 22,39                    | 21,53                    | 25,29                                     | 27,94                          | 27,02                          | 27,81                          |
| $R = 2,5$  | 23,18                               | 30,05                    | 32,11                    | 31,21                    | 29,46                                     | 35,49                          | 33,66                          | 36,02                          |
| $R = 5$    | 27,24                               | 42,32                    | <b>44,10</b>             | 42,87                    | 33,26                                     | 44,73                          | <b>41,20</b>                   | 45,96                          |

Ranges of the obtained integer-valued control magnitudes:

$$R = 0,1 \rightarrow -15 \leq u_t^{I_{\varepsilon(C)/1/2/3}} \leq 25,$$

$$R = 1 \rightarrow -6 \leq u_t^{I_{\varepsilon(C)/1/2/3}} \leq 8,$$

$$R = 2,5 \rightarrow -4 \leq u_t^{I_{\varepsilon(C)/1/2/3}} \leq 5,$$

$$R = 5 \rightarrow -3 \leq u_t^{I_{\varepsilon(C)/1/2/3}} \leq 4.$$

Moreover, most values lie within the indicated ranges, whereas the “extreme” values are rare and do not appear on all trajectories. Even without this remark, it is evident that the difference is substantial. This fully corresponds to the significantly increased difference between the performance of continuous and integer-valued strategies shown in Table 3.

A valuable contribution to the illustration of the computations is provided by examples of the trajectories  $X_t$  in Fig. 1. They clearly show differences in the performance of continuous and integer-valued controls. However, it is not possible to visually demonstrate the differences between the integer-valued variants  $I_{1/2/3}$  in these plots. The nature of the controls  $u_t^{I_{\varepsilon(C)/1/2/3}}$  can be compared using the illustrations in Fig. 2. The conclusions drawn from the performed computations are presented in the Conclusion.

## 5. Conclusion

The main result of the performed experiments is an objective demonstration of the ability to stabilize the considered system using exclusively integer-valued controls. A qualitative assessment of the trajectories in Fig. 1 is unambiguous: the controls  $u_t^{(I_{1/2/3})}$  clearly demonstrate the ability to achieve the same objective as the continuous control  $u_t^{\varepsilon(C)}$ . The figures (visually), as well as the numerical data in Tab. 3 (formally), show that the controls are capable of steering the system to drift in changing directions determined by the chain. It is evident that control performance deteriorates in the model with indirect observations, when the chain state is unknown. It is also evident that, as the penalty on the control input increases, the “magnitude” of the control decreases, the integer-valued strategy becomes more constrained (fewer admissible values), and increasingly underperforms the continuous strategy. However, as the penalty increases, the model as a whole becomes overly artificial and less representative of practice. The system becomes more concerned with avoiding control expenditure than with stabilization itself. One may note that, while in the first example with  $R = 0.1$  the quality of continuous control in the full-information and indirect-observation models differs by almost a factor of two, in the example with  $R = 5$  the difference is only 20%. That is, the system becomes less concerned with the direction of drift than with

minimizing control effort. Clearly, with further increases in  $R$ , one arrives at the situation where it is preferable not to control the system at all.

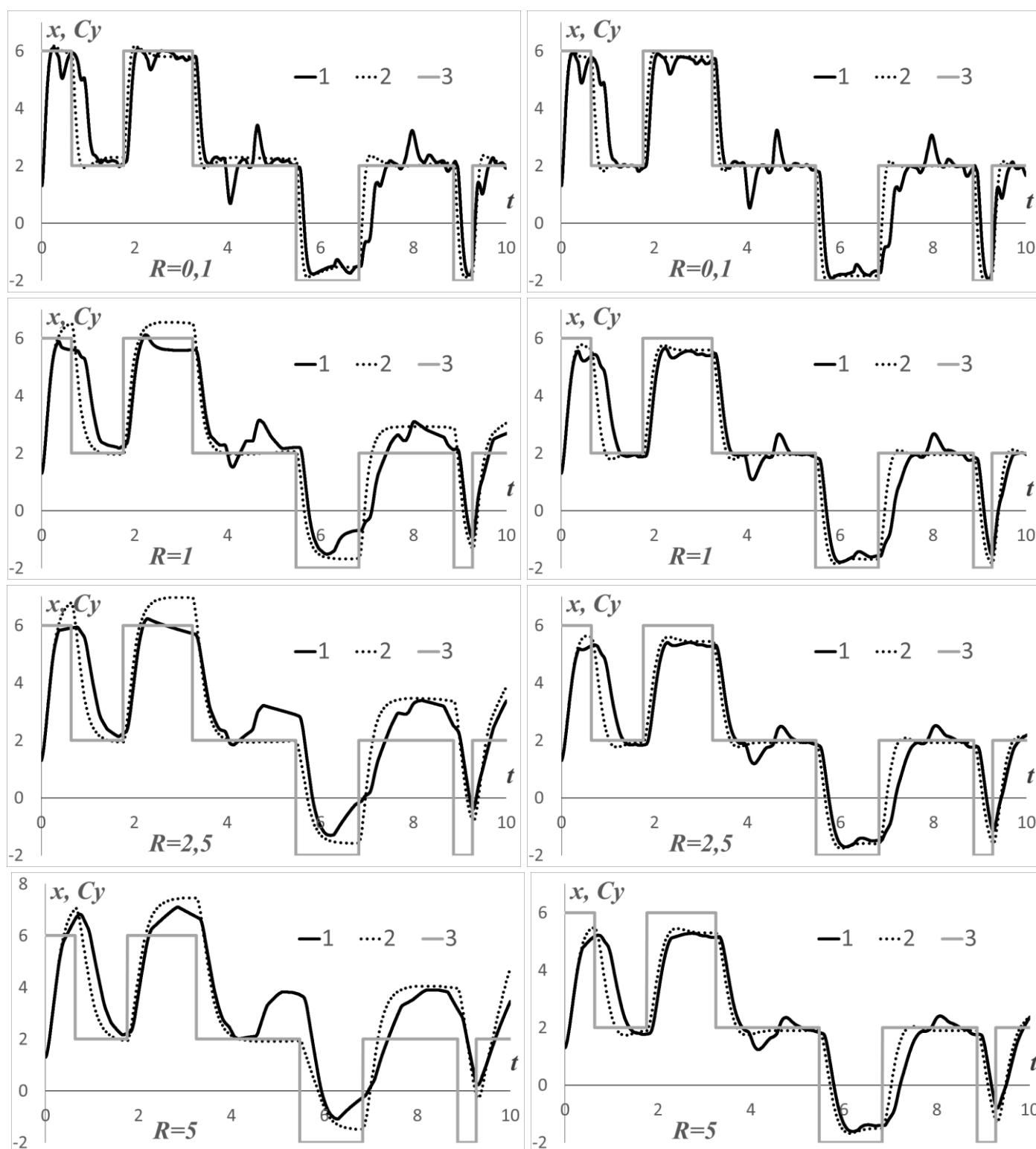


Fig. 1. Typical actuator trajectories: (left) 1 –  $U_{\hat{y}_t}^{\varepsilon(C)}$ , 2 –  $U_{\hat{y}_t}^{I_{1/2/3}}$ ,  
(right) 1 –  $U_{y_t}^{\varepsilon(C)}$ , 2 –  $U_{y_t}^{I_{1/2/3}}$ , 3 –  $Cy_t$  – направление дрейфа

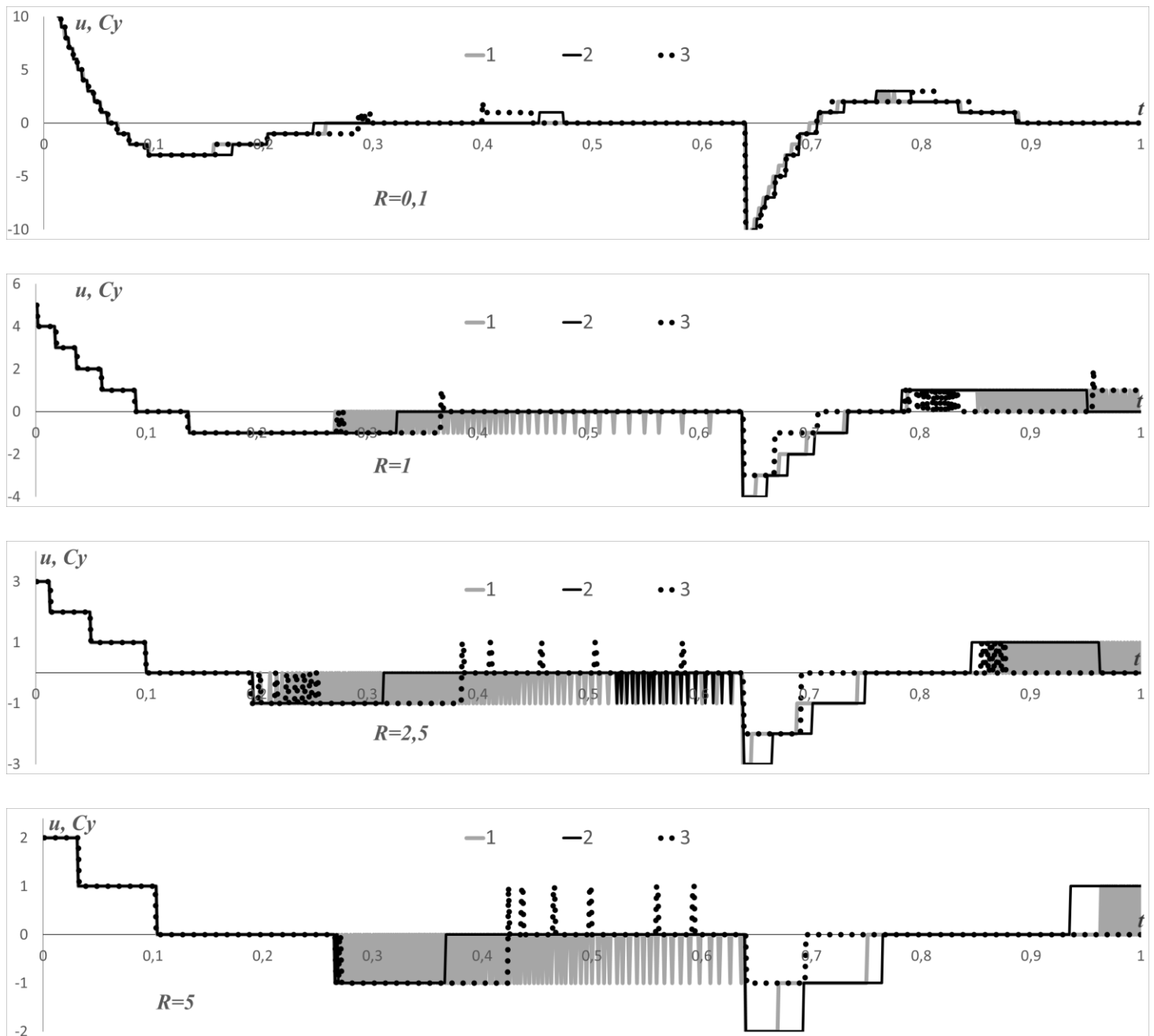


Fig. 2. Control trajectories: 1 –  $U_{y_t}^{I_1}$ , 2 –  $U_{y_t}^{I_2}$ , 3 –  $U_{y_t}^{I_3}$

In addition, while continuous control still manages to cope with the task to some extent in the example with  $R = 5$ , this example appears to be already borderline for integer-valued control. In particular, the quality of control  $I_2$  in the full-information model turns out to be worse than in the indirect-observation model. One may attribute this to the fact that this is a computer simulation with a relatively small sample size. Moreover, the controls  $I_{1/2/3}$  are heuristic and are not expected to provide guarantees. Nevertheless, a “deviation from the expected norm” is present, and this is a fact.

At the same time, as  $R$  increases, it becomes possible to more clearly illustrate the differences in the nature of the controls  $I_{1/2/3}$ . By sequentially examining Fig. 2, one can observe that initially these controls hardly differ. Subsequently, the simplest variant  $I_1$  demonstrates a tendency toward excessive “chattering”: the number of switchings increases, while there is no effect on the final performance. Finally, the last example conclusively shows that the most inert control is  $I_2$ , whose switchings are the most “cautious.” However, this has no direct impact on control performance, as expected.

Thus, overall, we have several viable strategies capable of replacing the optimal continuous control in the considered stabilization problem when integer-valued constraints are imposed on admissible controls.

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